

ACTIGAS

software – Export version

User manual



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1. GAS ANALYSER

1.1 INTRODUCTION

- The gas analyser is a PC based four (plus optional NOx) gas analyser.
- The gas analyser is intended for the maintenance and inspection of spark ignition engines.
- Gas data is transmitted from the gas analyser to the PC (or Laptop) for processing and printing.

1.2 SAFETY PRECAUTIONS

- a) The apparatus meets requirements for operation according to the EN 61010-1.
- b) The device shall be powered using only the power supply provided.
- c) Removing covers from the apparatus is not allowed, as this would cause an electric shock risk.
- d) **The device must be operated higher than the withdrawal probe inlet** due to the risk of the analyser becoming filled with a large quantity of water, coming from condensate in the exhaust system.
- e) **The device is intended for operation in the vertical position only. It should in no case be tilted even when switched off.** The reason is the danger that moisture condensed within the rough filter **F1** can flood the sensitive elements within the analyser.
- f) For emission control working sites and for service working sites, relevant safety regulations should be applied. The operator must organise appropriate safety training courses for service staff of the measuring apparatus. This can include training for working on vehicles while the engine is running.

Therefore, safety precautions must be observed according to the manufacturers of the car and of the engine, that are valid for testing the engine while it is running. Sufficient ventilation must be provided for exhaust fumes and outlet gases from the analyser.

- g) The device should be protected against ambient water in any form (rain, sprayed water, etc.)

1.3 DESCRIPTION OF DEVICE

The gas analyser is a four-part (optional five-part) analyser of exhaust gases, to be used for measurement and adjustment of petrol, LPG, and CNG engines, and for controlling emissions of those engines according to relevant national regulations. It can measure HC, CO, CO₂, O₂, and NO_x concentrations, and calculate CO_{cor} and the coefficient of air redundancy λ from those values.

1.3.1 FRONT PANEL

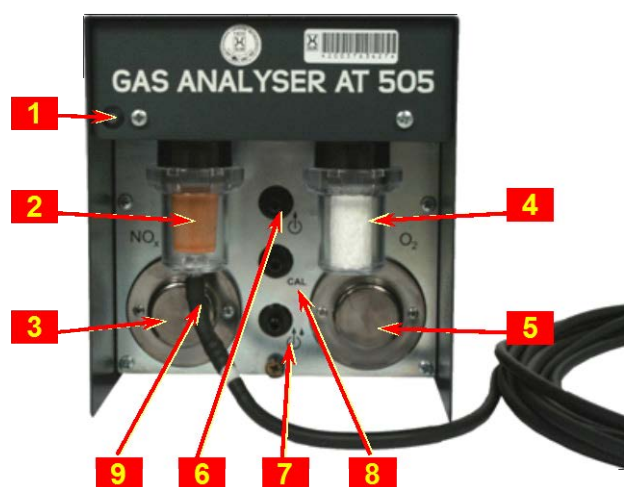
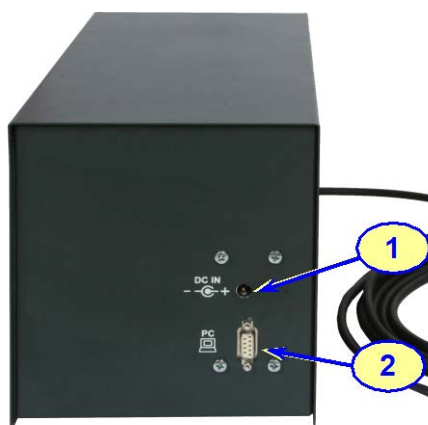


Fig. 1 – View of the front panel of the Gas Analyser

- 1 - Status LED:
OFF = the power supply is not connected, or the Gas Analyser AT505 software on the PC is not running and communicating with the Analyser.
Red = the power supply is connected, but the communication cable is not connected to the PC or the PC is not switched on
Orange = the analyser is ON, but inactive, heating up may be in process or it may be indicating a fault
Green = the analyser is ready to use
- 2 - Coarse filter (F1)
- 3 - Nitrogen sensor (NOx) (optional)
- 4 - Fine filter (F2)
- 5 - Oxygen sensor (O2)
- 6 - Exhaust gas outlet
- 7 - Condensate outlet
- 8 - Calibration gas inlet
- 9 - Exhaust gas intake

1.3.2 BACK PANEL



- 1 - Connector (Jack) for the power supply plug
2 - Connector for the computer serial cable

Fig. 2 – View of the back panel of the Gas Analyser

1.4 CONSTRUCTION OF APPARATUS

1.4.1 MECHANICAL PART

The device frame is fabricated from steel with a zinc finish, and is coated with a two-compound acrylic paint. The device case is also fabricated from steel with a zinc protective finish, and sprayed with a wear resistant powder coated paint. The exhaust probe is made of stainless-steel flexible hose.

1.4.2 ELECTRONIC PART

The internal electronic system and all clamps, connectors and sensors accessible from the outside are galvanically isolated, any voltages on them do not exceeding safe levels. The electronics consist of a printed circuit board, the electronic system of the optical testing bench, and the fan. In the case of the extended version including engine speed and engine temperature measurement, an additional circuit board is included for these sensors.

1.4.3 PNEUMATIC PART

This consists of a double membrane pump with two pumping channels. The **P1** channel pumps the exhaust gas to be measured from the exhaust probe to the optical bench and oxygen sensor (or nitrogen sensor NOx), and the **P2** channel continually empties condensate from the rough filter **F1**.

The solenoid enables zeroing of the measuring apparatus by switching from the exhaust probe to the zero setting inlet (which draws in surrounding air). The zero setting inlet includes an active carbon filter **F4**, which decreases hydrocarbons which may be present in the surrounding air.

The gas sample measured is subject to double filtration before entry to the optical bench. The first filtration stage is through a rough filter **F1**, which is made from sintered bronze grain. The second filtration stage is the fine filter **F2**, which is made from paper. Inside the apparatus there is safety filter **F3** that protects the solenoid, optics of the measuring bench, and the oxygen sensor (or nitrogen sensor NO_x) in case of incorrect fitting or failure of the external filters.

The apparatus is also equipped with a calibration inlet **CAL**, which is used for the connection of the calibration gas. The pneumatics include a vacuum switch, which is used for indication of decreased flow through the exhaust probe from taking in water or blockage, this is also used for the probe leakage test.

The device is equipped with automatic compensation for ambient air pressure.

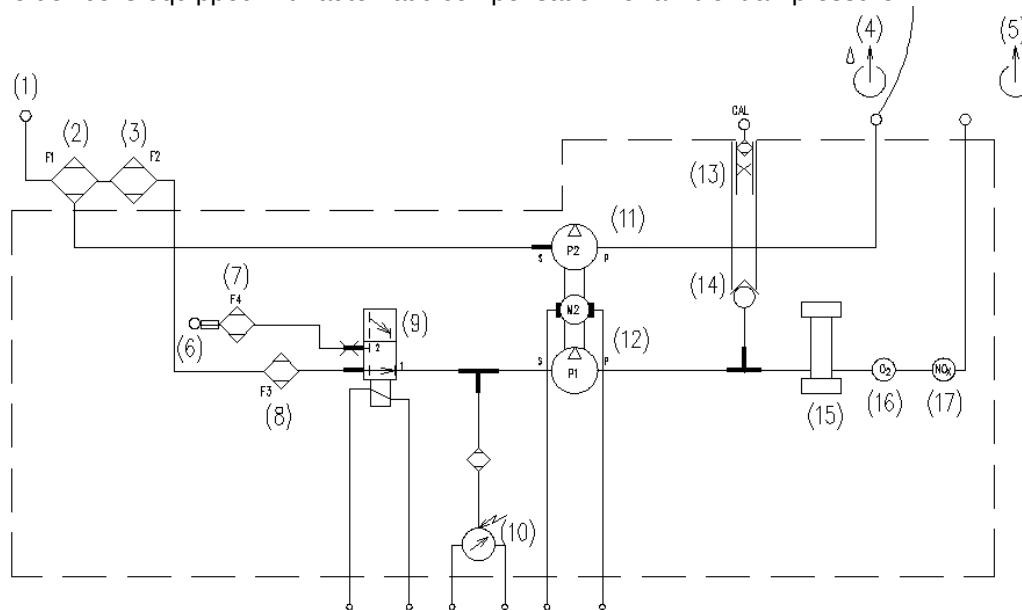


Fig. 3 – Scheme of pneumatic part

- | | |
|---------------------------------|---|
| 1- Exhaust gas intake | 10- Vacuum switch |
| 2- Coarse filter (F1) | 11- Condensate pump (P2) |
| 3- Fine filter (F2) | 12- Main pump (P1) – measured gas |
| 4- Condensate drain outlet | 13- Pressure reduction – inlet of calibration gas with pressure reduction |
| 5- Exhaust gas outlet | 14- Check valve |
| 6- Air intake (for zeroing) | 15- Optical bench |
| 7- Activated carbon filter (F4) | 16- Oxygen sensor (O ₂) |
| 8- Fine safety filter (F3) | 17- Nitrogen sensor NO _x (optional) |
| 9- Solenoid | cal- Calibration inlet |

1.5 MEASURING PRINCIPLE

The measuring principle for CO, HC, CO₂, O₂, and NO_x complies to the requirements **OIML R 99 class 0**. In the case of CO, HC, and CO₂, it is through non-disperse light attenuation measuring in an infra-red area (NDIR). Concentration of the three gases is measured in separate measuring cells. Signals obtained from the detectors are then compared with a reference signal. Results are compensated for temperature and barometric pressure changes.

O₂ and NO_x concentration measurement is done with electro-chemical sensors. The service life of these sensors is limited, and depends on the duration and level of concentrations, particularly of HC and Pb (from lead petrol detergents), which they are exposed to. Normally the service life should be at least six months. The Gas Analyser automatically monitors the sensors and reports when have require renewal.

A diagram of lambda dependency of individual gas compounds in exhaust gases is shown in Fig. 4. The richness of the fuel mixture is expressed by air redundancy coefficient λ . This diagram illustrates model emission behavior of a non-catalyst engine.

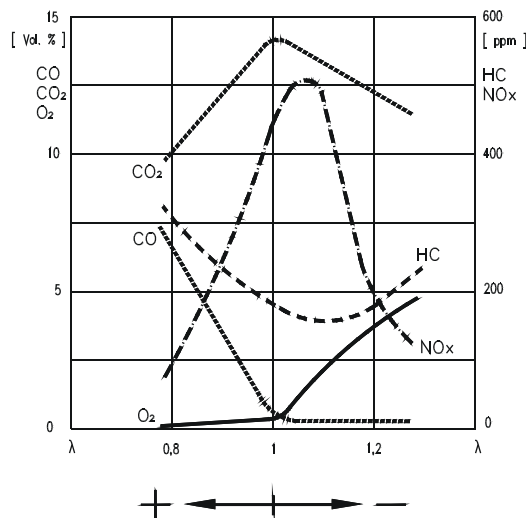


Fig. 4 – Diagram of dependency of concentrations of individual gas compounds in exhaust gases (+ rich mixture, - weak mixture)

2. OPACIMETER

2.1 INTRODUCTION

- The method is based on measuring the attenuation of light beam in a cell filled with the measured smoke.
- Both steady-state and acceleration measurements are possible.
- The acceleration test (obligatory during the emission inspection) registers the idle and over speed revolutions, acceleration time and peak opacity.

2.2 SAFETY PRECAUTIONS

- The apparatus meets requirements for operation according to the EN 61010-1.
- The LCS 2004 must always be connected to the network voltage, supplied by separable feeding Euro
- Removing covers from the apparatus is not allowed, there is danger of injury caused by electric current.
- For emission control working sites and for service working sites, relevant safety regulations are applied. In that spirit, the operator must organise training course for service staff of the measuring apparatus. From the point of view of measuring provided on the car, those acts are usual service actions, which are, however, executed at run.
Therefore, safety precautions must be kept according to the manufacturers of the car and of the engine that are valid for testing the engine during its run. Sufficient suction must be ensured of escaping exhaust gases, and outlet of outleting gases from the analyser, out of the measuring working site, too.
- Safety pins: network plug of sampling unit with fuse T 1.6 A
- The device should be protected against ambient water in any form (rain, sprayed water, etc.)**

2.3 DESCRIPTION OF DEVICE

Opacimeter is composed of personal computer (PC or Laptop not included in the set) used for displaying of measured values and communication with operator, and of the LCS 2400 sampling unit used for taking and processing of samples from oil engines exhaust gas.

The instrument's mechanical design is described in Chap. I - 2.4.1.

2.3.1 SAMPLING UNIT LCS 2400

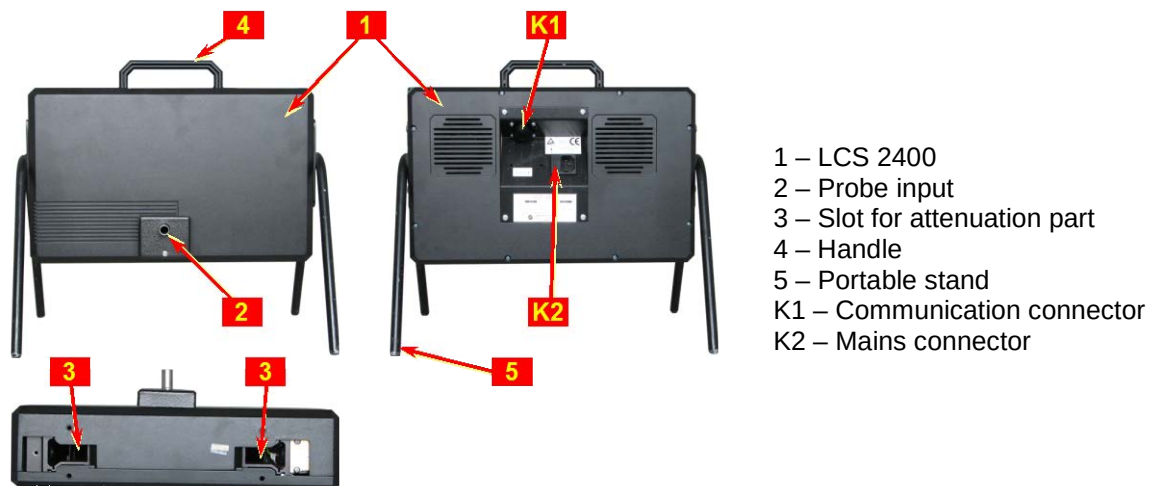


Fig. 5 – Sampling unit LCS 2400

The sampling unit is an integral part of the Opacimeter. It is connected to the PC via interconnecting cables, which are 7 meters long (connectors **K1**).

2.4 CONSTRUCTION OF UNIT

2.4.1 MECHANICAL PART

Coating of the sampling unit is made out of black matted plastic. Electronics of the sampling unit is situated in the sheet shading cover. Sampling probes are made of flexible stainless hose, provided with protective braiding. Prolonging hoses of probes are made of silicone rubber, resistant against crude oil products and increased temperature.

The stand for LCS 2400 is made of steel material and is sprayed with acrylate two-component paint.

2.4.2 ELECTRONIC PART

Inside electronic system and all clamps, connectors and sensors accessible from outside are galvanically separated from the network, and there is voltage on them not exceeding the safety voltage.

The sampling unit is fitted with electronic control of opacimeter optical system. Communication with the PC goes through serial line RS 232C.

Sampling unit's electronics is placed on two printed wiring boards, namely the microprocessor, memory & decoder board and board of alternating voltage distribution. Alternating voltage 230 V feeds both two built-in fans and heating system of measuring cell and detector.

2.4.3 PNEUMATIC PART

Pneumatic part the Opacimeter is placed in the sampling unit and is represented on the Fig. 6. Basis of the sampling unit is measuring cell (1), light source and detector block (2), with support of which the opacity measuring itself is processed. Mirror (3) serves for the reflection of the light, which is sent from the light source. Fans (4) direct the direction of smoke, so that no contamination of optic system occurs. Heating coil (spiral) (5) serves for the measuring cell warming up.

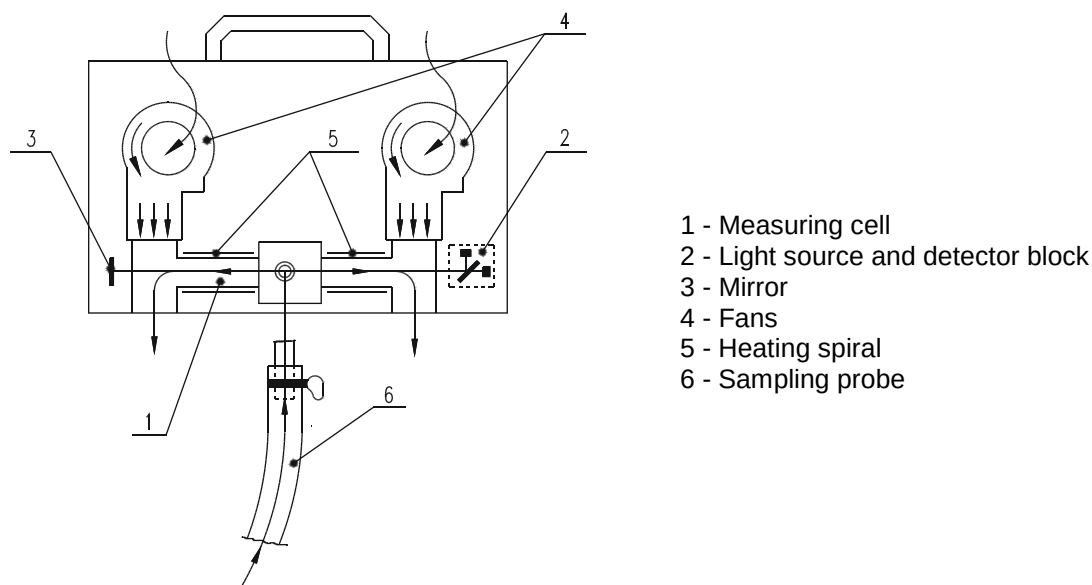


Fig. 6 – Pneumatic part of the sampling unit

2.5 MEASURING PRINCIPLE

The Opacimeter's smokiness measuring method is in accordance with ECE R24 directives for measuring by free acceleration method.

Principle of opacimeter measuring consists in measuring of amount of light passing through a sample of an exhaust gas. Absence of smoke in the measuring cell means zero opacity (optical density). Complete shading of the measuring cell (non-transparent gas) means 100 % opacity. As shown on Fig. 6, exhaust gas is led by sampling probe (6) to the measuring cell (1) with 364 mm effective length. Smoke column in the cell is measured by means of an optical system, where a green LED is used as a light source, which emits visible light of 560 nm. GaAs photodiode is used as a detector, which is heated to 45°C. For stability and accuracy purposes, the measuring cell is also heated to 75 °C.

The free acceleration method is a smokiness test, when engine changes its speed from idle to over speed with accelerator at full-throttle, and the engine is loaded by its own inertial mass and by clutch and gears at geared neutral.

During free engine acceleration (caused by setting of its accelerator from position „idle speed“ to position „full-throttle“) the opacimeter tests opacity in the given time with sampling time interval of 20 ms. Measured samples are mathematically evaluated and the front panel display shows resulting opacity $k [m^{-1}]$. This procedure repeats so many times as prescribed until measured values of smokiness fulfil the reproducibility condition. Resulting smokiness in absolute units $[m^{-1}]$ is calculated as arithmetic mean of last valid readouts.

1. GAS ANALYSER

1.1 DEVICE ACTIVITIES AFTER PROGRAM START-UP

1.1.1 DEVICE INITIALISATION AND OPTICAL BENCH HEAT-UP

After start-up, the initialisation of measuring devices and optical bench heat-up will take place. Until the bench is heated-up to its operating temperature, a warming up message will flash in the info box

1.1.2 LEAKAGE TEST - OPTION

As soon as the optical bench is warmed up to the required temperature, the leakage test will be carried out. **Before the test is started, the probe should be removed from the exhaust** and the red seal should be fitted.

The leakage test is carried out automatically once a day. To proceed with the leakage test follow the instructions on the screen.

1.1.3 OPTICAL BENCH RESET

This is executed automatically when the program starts. When the analyser is on it carries out a reset at intervals of 15 to 40 minutes, so that the required measurement accuracy is maintained. Resetting is not carried out in the course of a measurement.

During resetting air is drawn through a carbon filter to eliminate any hydrocarbons which may be present.

1.2 MAIN SCREEN DESCRIPTION

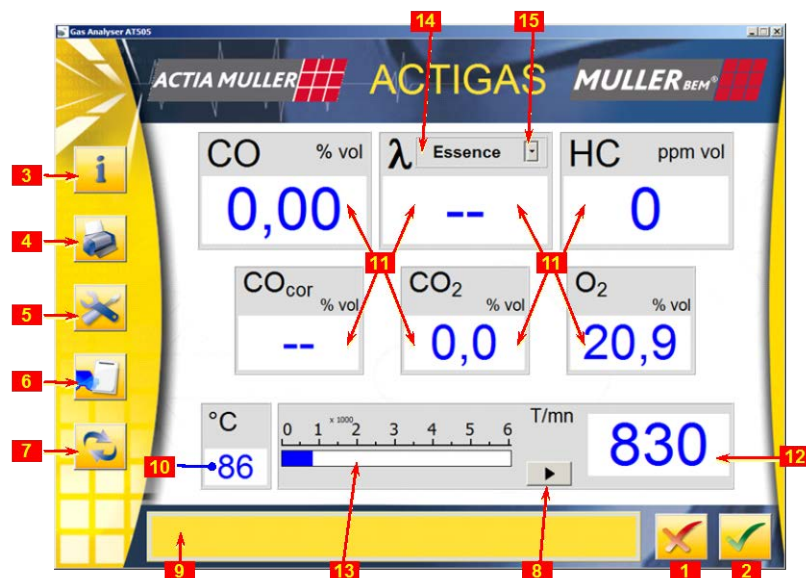


Fig. 7 – Initial Gas Analyser Software screen description

- 1 - **Button:** Used to exit from the program
- 2 - **Button:** Green = measurement enabled
Red = standby, fault or initialisation - **see the Note below**
by pressing this button you will switch between standby and measuring mode
- 3 - **Button:** Display information about the program and connection
- 4 - **Button:** Prints the currently displayed data
- 5 - **Button:** Configures several program parameters (see Fig. 8)
- 6 - **Button:** Used to start the emission test

- 7 - **Button:** Switches over to another application
- 8 - **Button:** Calibrate the Universal Speed Sensor. For information on how to read data using the vehicle ECU
- 9 - **Screen:** Displays information about program operation
- 10 - **Screen:** Displays the engine temperature
- 11 - **Screen:** Displays gas components
- 12 - **Screen:** Displays the engine speed
- 13 - **Screen:** Displays the engine speed graphically
- 14 - **Screen:** Selected fuel
- 15 - **Quick option:** fuel selection for lambda calculation (petrol, LPG, or CNG)

Note to Button 2 - Fig. 7 (**ENTER** Button):

The **ENTER** button is used for switching to emergency mode, where the pump is turned off. The apparatus can also move to emergency state also automatically. After transferring to the emergency state the apparatus always carries out zeroing.

1.3 INITIAL ANALYSER SETUP

The analyser settings can be changed to configure it as required.

Here you can change the setting of the Speed and Temperature Sensor, fuel type and several other features of the program. You can get to the setup mode by clicking on the button **F3** (pos. 5 - Fig. 7).

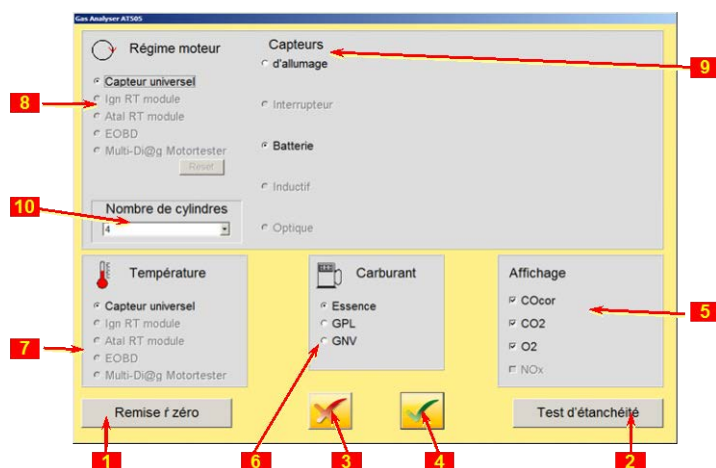
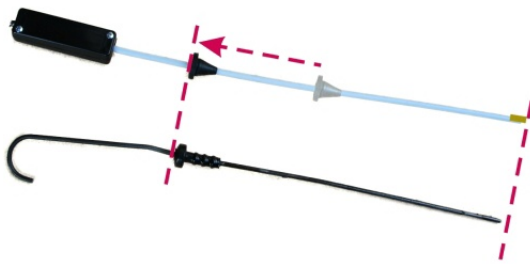


Fig. 8 – Parameter setting of the program

- 1 - **Button:** Start optical bench reset
- 2 - **Button:** Start leakage test
- 3 - **Button:** Exit setup without saving changes
- 4 - **Button:** Exit setup and save changes
- 5 - Selection of gases to be displayed
- 6 - Selection of fuel for which the lambda coefficient will be calculated
- 7 - Selection of sensor from which the engine temperature will be read
- 8 - Selection of sensor from which the engine speed will be read
- 9 - Setup of individual probes
- 10 - Setup of number of cylinders or of engine type (2 stroke, 4 stroke)

1.4 OIL TEMPERATURE MEASURING

1.4.1 OIL TEMPERATURE SENSOR



The proper oil temperature sensor type (with correct length set) should be used. After removing oil dip stick, set the thermometer's rubber stop (with help of the dip stick) so that the thermometer is about 20 mm deep (30 mm max.) in the oil.

1.5 ENGINE SPEED MEASURING

1.5.1 UNIVERSAL SPEED SENSOR

The measuring is carried out by connecting a suitable cable to the battery of the vehicle. There are two options to select from:

- Connect it to the auxiliary power outlet (cigarette lighter socket)
- Connect it to the vehicle's battery terminals: Red terminal is + pole; Black terminal is – pole

1.5.2 RECOMMENDATIONS

- It is advantageous not to have fully charged battery - switch the headlights on for a while before measurement.
- While the measurement is in progress try to increase the current consumption (e.g. switch headlights and/or other electrical instruments on).
- Direct connection to the battery terminals usually gives more reliable results than other points.
- In case of implausible speed readings try to resynchronise the sensor by pressing the **F9** key.

1.6 PRINTING VALUES

The print queue behaves like a storage folder for collecting individual records of the measurement results to be printed later. The contents of the print queue are printed when the user desires.

Press the **Print** button to add the screen data to the print queue. An information window will be displayed

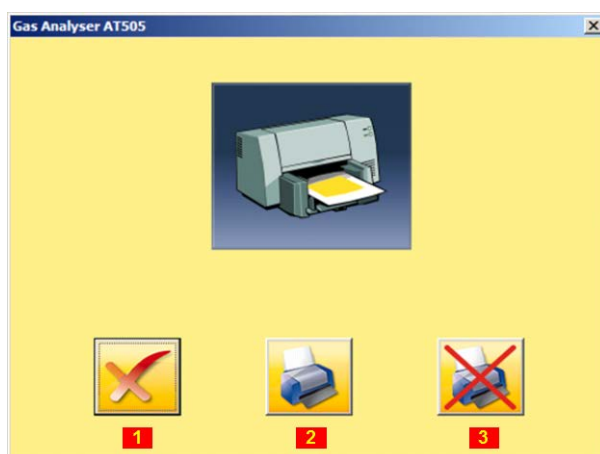


Fig. 9 – The printer options are displayed

- 1 to continue measuring – the existing saved data is preserved
- 2 prints all data from the print queue to the printer
- 3 deletes all data in the print queue (without printing)

Note:

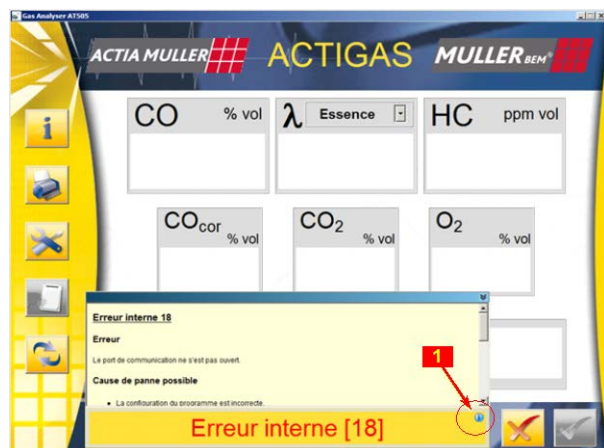
The print queue is automatically deleted on exiting the program.

1.7 EMISSION STATION SYSTEM INFORMATION

If you press the **Info** button, information about software, version number, calibration date etc. is shown.

1.8 ERROR ALERTS BY THE ANALYSER

All errors and other information are displayed on the PC screen. Test data can be printed out with the printer. On the gas analyser there is a status LED.



If an icon (see pos. 1) is displayed in the information box, click it to display a panel with further information. Clicking again removes it.

Fig. 10 – Example of an error message

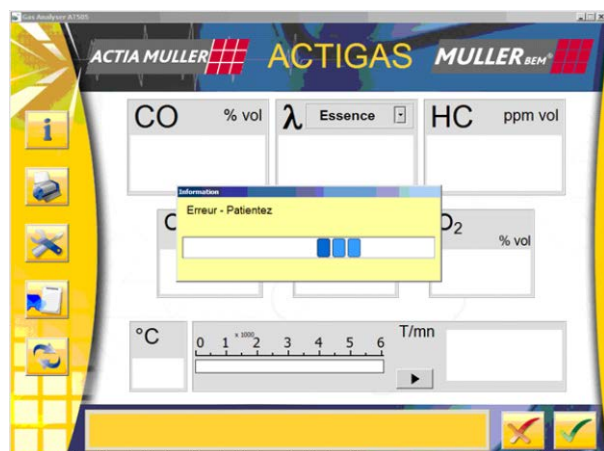
1.9 MAINTENANCE

The apparatus is designed to be as maintenance free as possible. Special attention should be given to protecting the analyser inlet from undue contamination. The exhaust probe should not be left on the floor or any other contaminated area, where it can suck in dust, contaminants, or water. The probe should only be inserted into the exhaust when required during emission measuring, and removed when not required.

Calibration of the gas analyser should be carried out every 6 months by an approved calibration service.

1.9.1 MEASURES TO AVOID WATER CONDENSATION IN OPTICAL BENCH

In case of drop in intake probe pressure the device reports an error condition (see Fig. 11).



The condition is a result of a choked or blocked intake probe (hose) or sucking water.

Fig. 11 – Blocked probe warning

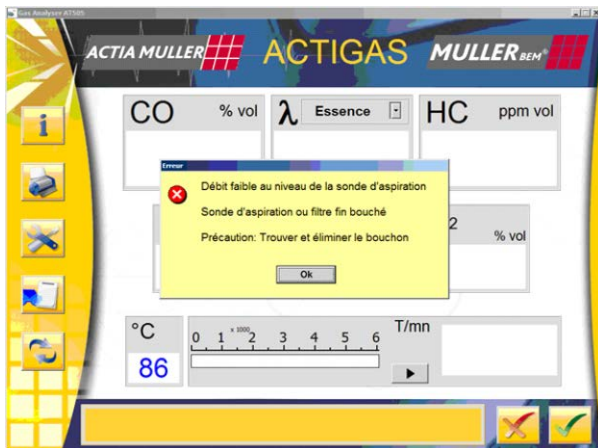


Fig. 12 – Error message with request for a fix

The device operates the valve several times in order to reduce under pressure (clicking sound may be heard from inside).

The user is informed with the error message and asked for fixing the problem.

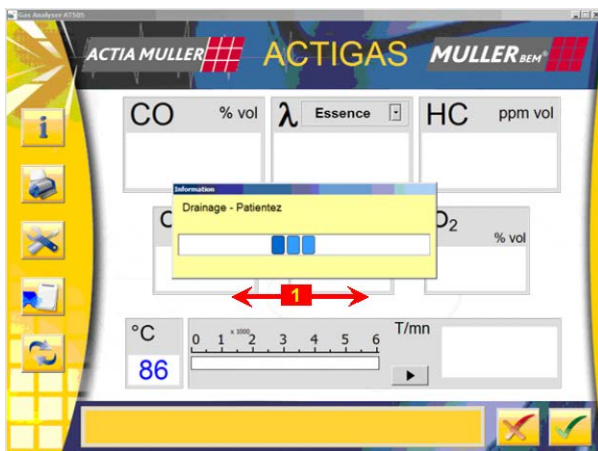


Fig. 13 – The display during the probe cleaning phase

The following action of the analyzer tries to avoid water penetration into a sensitive optical bench. The device actuates the valve and the pump in several cycles (with corresponding sound effects) and in the meantime it is testing if conditions returned back into a normal state, i. e. the probe is clean again. In case of success the measurement can continue.

1.9.2 COARSE FILTER INSPECTION AND CLEANING

The sintered bronze grain coarse filter may be cleaned by washing in spirit. After cleaning the filter it must be rinsed out thoroughly in clean water and blown through with pressurised air.

Filter contamination can prolong the residual hydrocarbon check and so increase the overall time of the emission test.

1.9.3 FINE FILTER INSPECTION AND RENEWAL

If the paper filter is contaminated it must be replaced with a new one. Its lifetime depends on the emission levels of the vehicles tested. Where the engine is equipped with a catalyst, the life service of the filter is minimally 50 emission tests. With engines that are in a poor condition, and in the case of two-stroke engines, the life service of the filter may only be 5 emission tests.

Filter contamination can prolong the residual hydrocarbon check and so increase the overall time of the emission test.

1.9.4 OXYGEN SENSOR REPLACEMENT

The oxygen sensor is accessible from the front panel of the device after the cover of the oxygen sensor is removed, see Fig. 1, Pos. 5.

Procedure for oxygen sensor replacement:

- Disconnect the gas analyser power supply.
- Remove the oxygen sensor cover.
- Disconnect the oxygen sensor connector noting its orientation.
- Unscrew the old oxygen sensor.
- Screw in the new oxygen sensor – do not over-tighten it.
- Fix the oxygen sensor connector in place observing correct orientation. Incorrect orientation will cause an error message to be displayed when measurement is attempted
- Refit the oxygen sensor cover.
- Reconnect the power supply. Carry out an emissions test to check the replacement oxygen sensor is working correctly.

1.10 TECHNICAL SPECIFICATION

Quantities with accuracy figures provided are guaranteed. Other values are for information only.

1.10.1 FUNCTION PROPERTIES

Measurement	Range	Resolution	Accuracy	Note
CO	0 - 10 % vol	0.01 % vol	0.03 % vol or 5 % RV	
CO ₂	0 - 20 % vol	0.1 % vol	0.5 % vol or 5 % RV	
HC _(hex)	-12 - 2000 ppm vol	1 ppm vol	10 ppm vol or 5 % RV	
	2001 - 9000 ppm vol	10 ppm vol	or 5 % RV	
O ₂	0 - 4 % vol	0.01 % vol	0.1 % vol or 5 % RV	
	4 - 21 % vol	0.1 % vol	5 % RV	
CO _{cor}	0 - 10 % vol	0.01 % vol		1
NO _x	0 - 5000 ppm vol	1 ppm vol		
Lambda	0.500 - 2.000	0.001	ISO 3929	2
RPM	400 - 2000 min-1	10 min-1	20 min-1	3
	2001 - 9999 min-1		2 % RV	
Temperature	0 - 150 °C	1 °C	2°C	3

RV = of reading value

Note no.:

1 - Calculation: at $CO + CO_2 \leq 15\% \text{ vol}$ $CO_{cor} = \frac{15 \cdot CO}{CO + CO_2}$
at $CO + CO_2 > 15\% \text{ vol}$ $CO_{cor} = CO$

2 - Calculation: Brettschneider's equation **)

3 - Only for the version of analyser with built in engine speed and temperature measurement facility.

**) CALCULATION OF AIR REDUNDANCE COEFFICIENT λ

The value is calculated from quantities of exhaust gases measured, according to the Brettschneider (BOSCH - Technical report, vol. 6 (1979), No. 4, page. 177-186). Relation for coefficient λ is:

$$\lambda = \frac{[CO_2] + \frac{[CO]}{2} + [O_2] + \left\{ \frac{H_{cv}}{4} \cdot \frac{3,5}{3,5 + \frac{[CO]}{[CO_2]}} - \frac{O_{cv}}{2} \right\} \cdot ([CO_2] + [CO])}{\left(1 + \frac{H_{cv}}{4} - \frac{O_{cv}}{2} \right) \cdot ([CO_2] + [CO] + K_1 \cdot [HC])}$$

where [] means concentration of elements in % of volume
K₁ re-calculation factor FID to NDIR measuring (= 6.10⁻⁴)
H_{cv} ratio of atomic numbers of hydrogen to carbon
O_{cv} ratio of atomic numbers of oxygen to carbon

1.10.2 GENERAL INFORMATION

Supply voltage.....	100 to 240 V AC / 47 to 63 Hz or 10 to 18 V DC
Power input	40 W max.
Warm-up time.....	10 min max. (at 25°C)
Communication interface	RS232
Weight	5 kg max.
Dimensions	330 * 178 * 195 mm (depth * width * height)
Operating temperature	5 to 40 °C
Operating relative humidity	up to 90 % non-condensing
Operating atmospheric pressure.....	860 to 1060 hPa
Storage temperature	-10 to 45 °C

1.10.3 STORAGE AND TRANSPORT

Storage temperature-10 to 45 °C

The gas analyser must not be transported or stored outdoors where it may be exposed to rain or other inclement conditions. Areas where it may be exposed to acids, chloride, mercury vapours, etc should be avoided.

1.11 CALIBRATION AND SERVICE

The analyser should be calibrated at the required interval by an approved calibration service.

2. OPACIMETER

2.1 PROGRAM STARTUP

After the program has been started, the heating-up test for the Sampling Unit measuring cells (LCS 2400) will be carried out. Until the unit is heated up to the required temperature (75°C), the message about the heating is blinking in the info box and the current measuring cell temperature is shown in the temperature box.

2.2 MAIN SCREEN DESCRIPTION

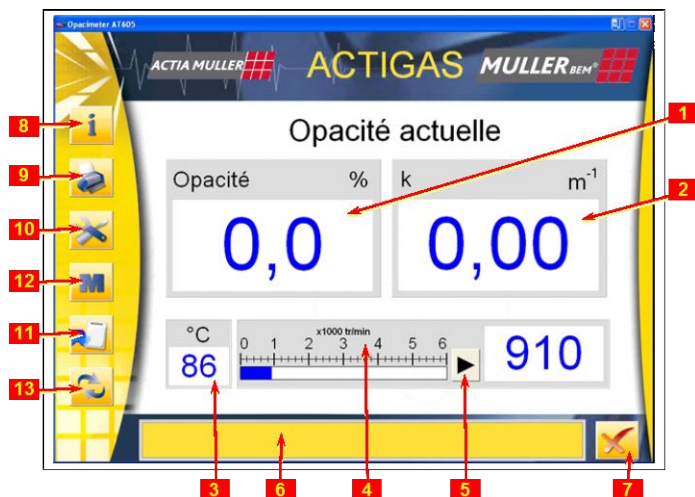


Fig. 14 – Main screen – Current Opacity Mode

- 1 - **Screen:** Current opacity value in percentage
- 2 - **Screen:** Coefficient of the current opacity value in units [m-1]
- 3 - **Screen:** Engine temperature (engine oil temperature)
- 4 - **Screen:** Box showing the engine speed both in figures and graphically
- 5 - **Button:** Used to synchronize the Speed Sensor
- 6 - **Screen:** Box of info and fault messages
- 7 - **Button:** Used to exit the program
- 8 - **Button:** Used to activate the "Program info"
- 9 - **Button:** Used to print the measured values
- 10 - **Button:** Used to enter the "tools" for setting the parameters etc.
- 11 - **Button:** Used to start the emission test
- 12 - **Button:** **M** – mode – Used to change the **Current Opacity Mode** to **Peak Opacity Mode** and vice-versa.
- 13 - **Button:** Used to start an external application e.g. Multi-Diag Scope

3. LINEARITY TEST AND CALIBRATION

As soon as the cell is heated up to the required temperature, the linearity test and calibration will be carried out – **before the test is started, the probe shall be removed from the exhaust.** If the probe is not removed from the exhaust, the calibration may be carried out incorrectly and error will be displayed.



Fig. 15 – Initialization (calibration + linearity test)

During this process the info box shows the “Initialization” text (Pos.1)

Then the program is automatically switched over to the “**Current Opacity**” Mode, see Chap. Fig. 15 in which the current values are measured.

Note:

Within the first 15 seconds or so after the initialization is completed, the speed and temperature values may be scored out (----) because of the initial synchronization of the Speed Sensor. Then the speed and temperature values shall appear.

4. INFORMATION ABOUT SOFTWARE VERSION

If you press the **Info** button or the **F1** key, the information about software, version number, LCS calibration date etc.

4.1 TOOL PANEL

You get in the Tool Menu by pressing the **Tools** button (Pos.10 - Fig. 14) or the **F3** key.

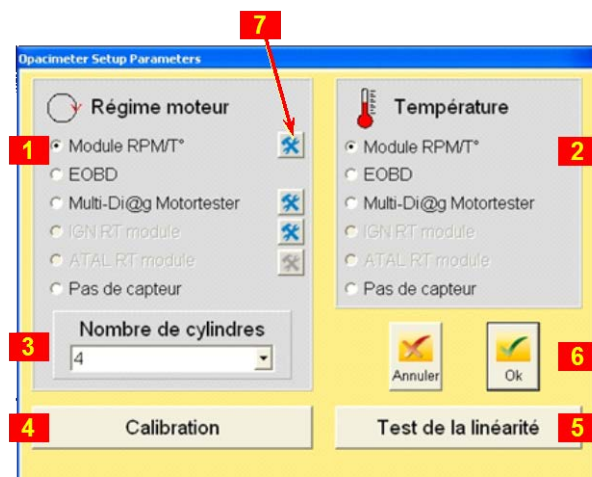


Fig. 16 – Tool panel main screen

- 1 - Setup of the Speed Sensor type.
- 2 - Setup of the Temperature Sensor type.
- 3 - Setup of the number of engine cylinders.
- 4 - Button used to perform the zero check of the Sampling Unit (LCS2400).
- 5 - Button used to perform the linearity test of the Sampling Unit (LCS2400).
- 6 - Save settings
- 7 - Button for Motortester settings - see Fig. 17

4.2 SETTINGS OF MOTORTESTER/MINITESTER PARAMS AND UNIVERSAL SENSOR SETTING

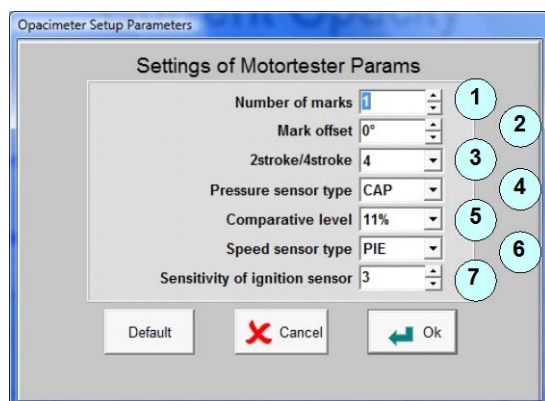


Fig. 17 – Settings of Motortester/Minitester Params

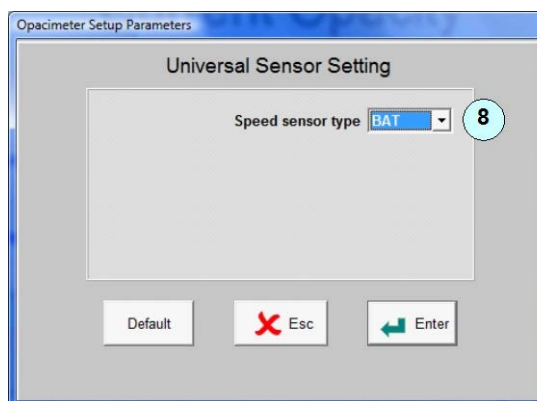


Fig. 18 – Universal sensor setting

Note to Fig. 17:

- (1) **Number of marks:** indicates the number of marks on the flywheel of the engine crankshaft. It can be set in the range from 1 to 255.
- (2) **Mark offset:** indicates the shift of the mark in from the top dead centre of the crankshaft in degrees. It can be set in the range $\pm 90^\circ$.
- (3) **2 Stroke/4 Stroke:** the engine type (2-stroke or 4-stroke). Note: from the ignition aspect, a four-stroke engine without ignition distributor and with a dual-output coil behaves as a two-stroke engine.
- (4) **Pressure sensor type:** (this parameter can **only be used with diesel engines**)
The pressure sensor CAPELEC = **CAP** is supplied as a special accessory with the Smokemeter. You can also use an **AVL** sensor. This option indicates the sensor type used.
- (5) **Comparative level:** (this parameter can **only be used with diesel engines**).
This is the level of the pressure sensor signal course according to which you shall set the preinjection amount. You can set 11% or 15% of the level from the maximum pressure signal level.
- (6) **Speed sensor type:**
 - a) Inductive (**IND**), semiconductor (**SEM**) or ignition (**IGN**) sensors can be selected for spark ignition engines.
 - b) Inductive (**IND**), semiconductor (**SEM**) or pressure (**PIE**) sensors can be selected for diesel engines.
- (7) **Sensitivity:** Here you can set the sensitivity of the **ignition** or **pressure** sensor input. It is recommended to change this value from the default value (=5) only when the engine speed measurement is significantly unstable. By changing this value you can eliminate this problem. A sensitivity of 1 to 7 can be set.

Note to Fig. 18 –:

- (8) **Speed sensor type:** (this parameter can **only be used with diesel engines**)
Battery (**BAT**) or ignition (**IGN**) sensor can be selected

4.3 TECHNICAL SPECIFICATION

We guarantee only the values where tolerances/limits are stated. Other values are only of informative character.

4.4 FUNCTION PROPERTIES

Measured quantity	Range	Resolution	Accuracy
Absorption coefficient (k)	0 – ~ 16 m ⁻¹	0,01 m ⁻¹	± 0,15 m ^{-1*} ; ± 0,30 m ^{-1**}
Opacity (N)	0 – 100 %	0,1 %	± 2 % absolute
Temperature	0 – 150 °C	1 °C	± 2 °C
RPM	400 – 2000 min-1 2001 – 9999 min-1	10 min-1	± 20 min-1 ± 2 % RV
Acceleration time	0 – 99,99 s	0,1 s	± 0,2 s
Acceleration measuring according to directive ECE R24			

RV = of reading value

^{*)} in range 0,0 to 2,5 m⁻¹

^{**)} in range 2,5 to 4,0 m⁻¹

4.5 GENERAL INFORMATION

Input Power (sampling unit)230 V AC +10% / -15%; 0,78 Amp, 50/60 Hz

Sample cell Temperature75 °C ±1°C

Effective Optical Path Length (EOPL).....364 mm

Startup time.....15 minutes @ 20°C ambient

Operation Temperature.....+5 to +40°C

Operation Humidity0 to 95% RH

Storage Temperature-32 to 50 °C

Communication interfaceUSB 2.0

Weight (sampling unit)4,5 kg max.

Dimensions (sampling unit).....235 x 380 x 90 mm max. (height x width x depth)

4.6 AMBIENT CONDITIONS

Operation temperature5 to 40 °C

Running moisture relativeup to 90 % non-shrinking

Electromagnetic compatibilityaccording to EN 50 081 - 1

4.7 STORING AND TRANSPORT

Storing temperature-10 to 45 °C

The instrument mustn't be transported or temporarily stored on open-air space, that isn't protected against rain, and in vehicles or areas, in which acids, chlorine, mercury vapours, etc. are stored.

5. PROBES AND SENSORS

5.1 SAMPLING PROBES

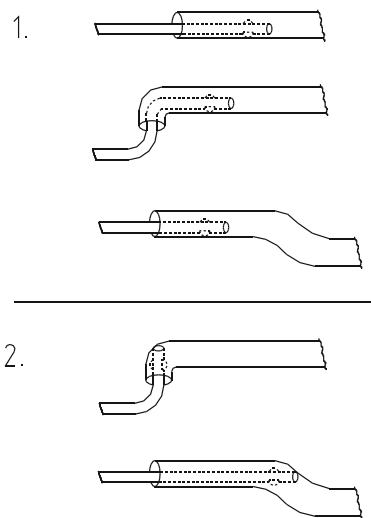


Fig. 19 – Sampling probe placing in exhaust of the car (1. - correctly, 2. - incorrectly)

1. STARTING APPLICATION

The starting application is designed for starting of other programs – Gas Analyser or Opacimeter. The starting application provides a communication with the PC center via GiegNet/GiegLan protocols and also starts configuration or manages stored protocols.

1.1 STARTING OF APPLICATION

This application is started automatically after start PC. The application can be started by clicking on the icon on the desktop (viz. Fig. 20) or standardly through menu Start.

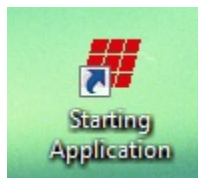


Fig. 20 – Application icon on the desktop

1.2 WORKING MODES

The application can be used in two modes.

- a) **Mode Controle Technique** – designed for the emission stations where the official tests are performed. This mode support the GiegNet/GiegLan protocols.
- b) **Mode Garage** – designed for service station where the devices are used only for diagnostic or engine parameters setting.

1.2.1 MODE CONTROLE TECHNIQUE



Fig. 21 – Main screen of Mode Controle Technique

- 1. GasAnalyser button
- 2. Opacimeter button
- 3. List of orders
- 4. Economy mode button
- 5. Garage mode button
- 6. Archive button
- 7. Button of info about programs
- 8. Configurator button

EMISSION TEST START

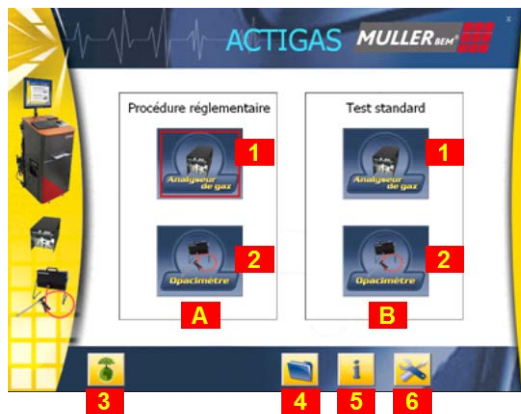
By selection from item list the right device is start (GasAnalyser or Opacimeter) automatically.

ES – petrol

GO – diesel

All available data are inserted into test (if the GiegNet or GiegLan is used). If you select the New car item, then you must select the right device yourself and the test will start with blank fields.

1.2.2 MODE GARAGE (START)



1. GasAnalyser button
2. Opacimeter button
3. Economy mode button
4. Archive button
5. Button of info about programs
6. Configurator button

-
- A. Group of official emission test
 - B. Group of service test

Fig. 22 – Main screen of Garage mode

A) Official emission test group

The buttons from this group start the official emission test. All In this mode the user can use the device for setting an engine of car or he can use the special tests.

B) Test standard group

The buttons from this group start the devices in a mode current value measuring. In this mode the user can use the device for setting an engine of car or he can use the special tests.

1.3 ARCHIVE (START)

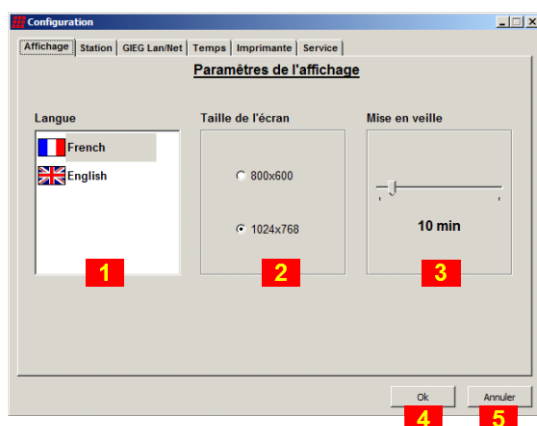
In archive there are stored the measured protocols for specify time interval. This time interval can be change in configurator. After expire this interval the protocols are removed (no restoring possible). In archive there is possible the protocols to view, print or save.

1. PROGRAM CONFIGURATION

In order to be able to perform the program configuration under Windows 2000/XP/Vista/7 you need to have the administrator access rights.

Configuration software is used for setting of parameters of emission devices – Starting application, Gas analyser and Opacimeter. The configuration utility is started by clicking on the icon Configuration or from menu: **Start / All softwares / Actigas / Configuration**. Some parameters can change only authorized person. It is recommended to close Gas analyser and Opacimeter application before configuration.

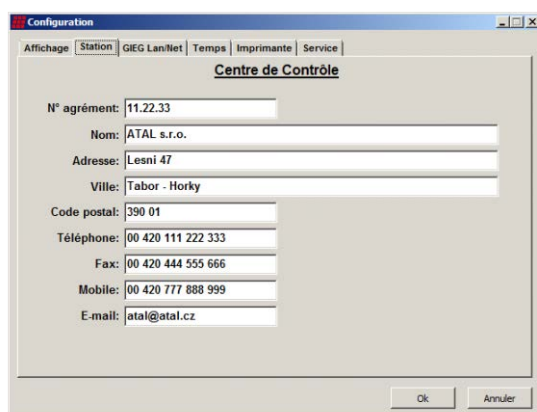
1.1.1 SCREEN SETUP



- 1 Language selection
- 2 Screen resolution
- 3 Time for stand-by mode (from 1 to 90 minutes)
- 4 Button Ok – save parameters and end of configuration
- 5 Button Cancel – end of configuration

Fig. 23 – Screen setup

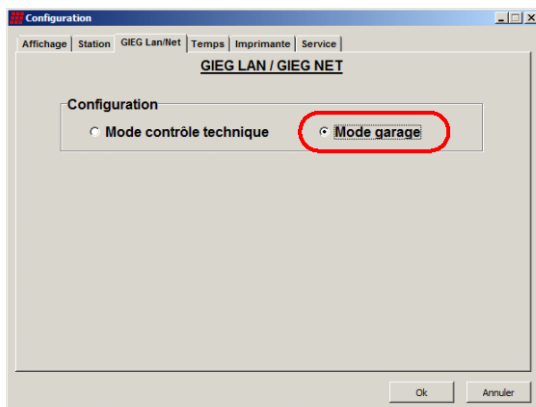
1.1.2 STATION



Identification of emission station in the printout.

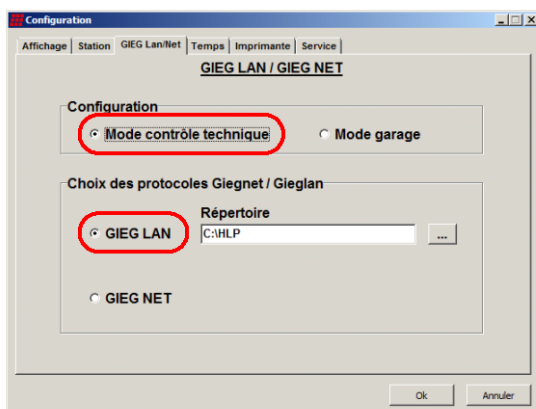
Fig. 24 – Station

1.1.3 GIEG LAN COMMUNICATION



„Mode garage“ is selected. Gieg protocol is not supported.

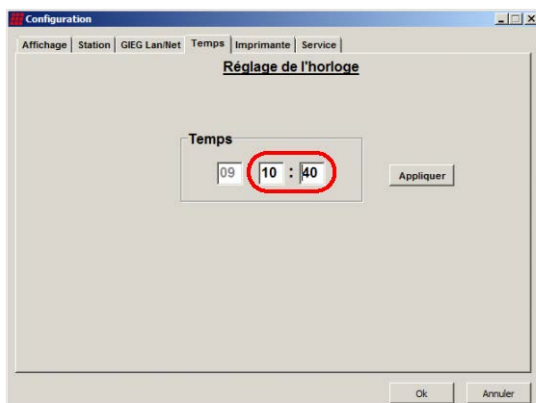
Fig. 25 – Garage mode selection



The protocol Gieg Lan is selected – it is necessary to choose the path for Gieg Lan communication. Example is: C:\HLP

Fig. 26 –Gieg Lan selection

1.1.4 TIME



Time modification.

It is possible to change only minutes and seconds. Time is changed by pressing to button “Apply”.

Fig. 27 – Time modification

1.1.5 PRINTERS

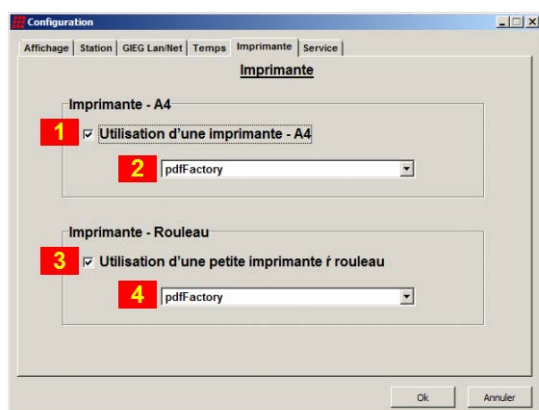


Fig. 28 – Printers

- 1 selection to use / not-use printer with A4 paper format
- 2 name of printer with A4 format
- 3 selection to use / not-use small printer with roll paper format
- 4 name of printer with roll format

If no printer is selected the software uses default printer of operating system.